

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time. Encode in orbit, decode in AI’s reasoning with emem.

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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)



TRY IT
Your phone becomes Agent B.
Resolve the record
and run its checks.



2.02 GB scene **84 characters**
The reference to this pixel's NDVI record fits in one text message.
84 characters; checking it fetches the 1,115-byte record and its log entry.

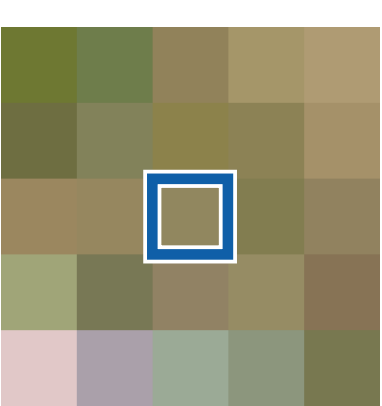
From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor Keep the source scene.	Location · product · time Lookup identity	Canonical record + CID Batch attestation	Pass the reference MCP · REST · A2A	Retrieve the same record Recover its provenance.	Hash · bind · sign · source If one fails: refuse.	Reason with cited evidence Carry the reference onward.

1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a satellite reading. A relay changes the evidence. Agent B acts on it or checks the reference.

Agent A cites	handoff	relay	Agent B	B acted on corrupted evidence agents, pooled Claude	declined	used the genuine record
 Agent A reads NDVI 0.4709 and cites it Sentinel-2A L2A, 25 Sep 2026 outlined: the 10 m pixel the record names RGB from recorded grids	A prose The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha... JSON { "about": { "field": "Keylong, Lahaul (32.57126 N, 77.03448... retrieved text (RAG) Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell ... opaque id record ref obs-7d1c44 emem reference emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk6...	relay or faulty signer changes: value cell time band source derivation stale state signature pixel	B reads the text	254 / 276	22	0
	B reads the fields	164 / 276	106	6		
	B retrieves a passage	220 / 276	56	0		
	B fetches the sender's record	154 / 288	88	46		
	B resolve re-hash place / time / band signature log recompute source re-read	0 / 300	264	36		

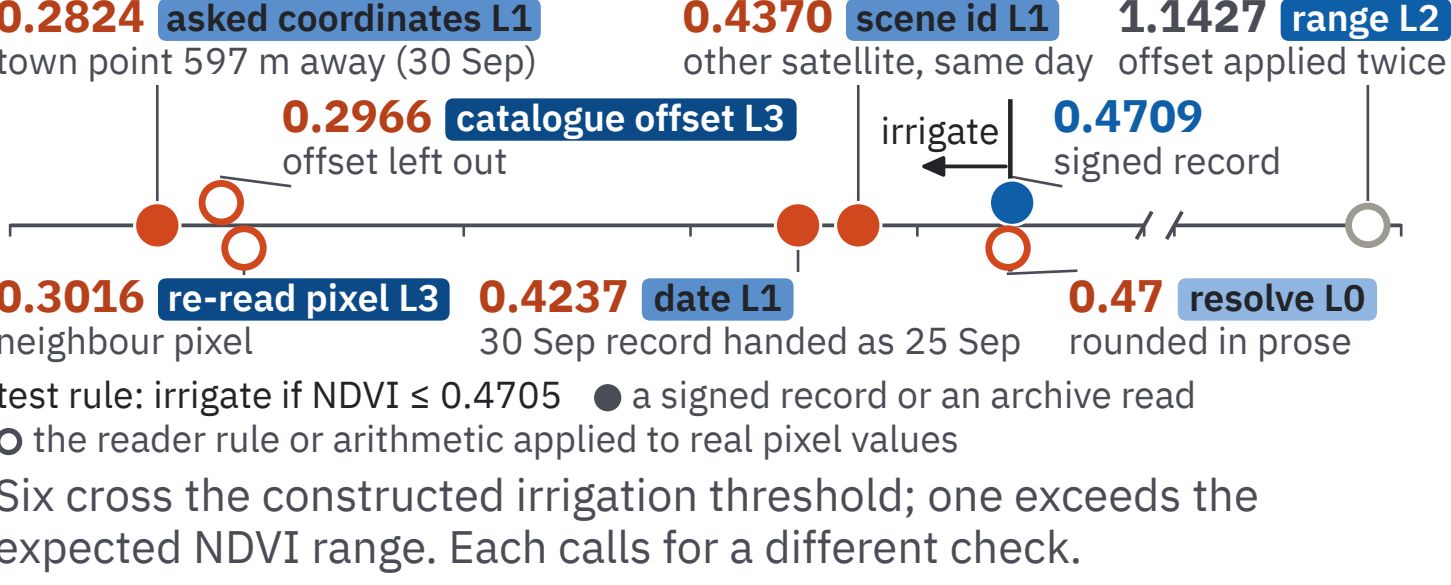
Across the 23 shared items, agents acted on 254 of 276 corrupted handoffs in prose, 154 of 276 with an opaque id, and 0 of 276 with an emem reference (2 of 276 without the instruction to check).

By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked into the AI’s reasoning (SAT-042 harness, not flown).

2 How does a value change between agents?

One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



3 Where did the evidence change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

Δz : change in readout. The terms list possible causes (environment, sensor, geolocation, encoder, residual), not measured shares.

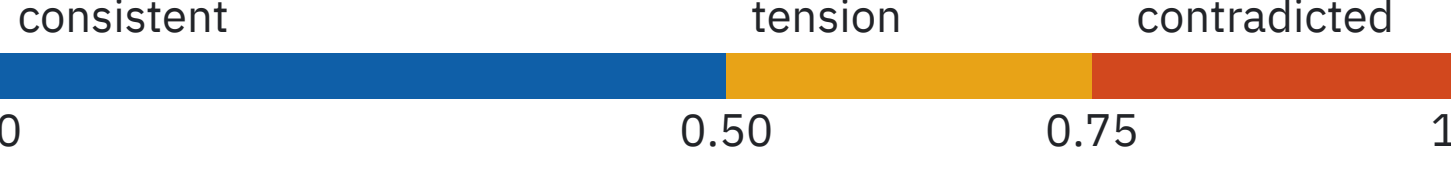
IMPLEMENTED HEURISTIC · DRIFT-ANCHOR SCORE

$$d = |x - a| \text{ for finite } x, a \text{ and } 0 < \sigma < \infty$$

$$r = d / (3\sigma); s = r / (1 + r)$$

Invalid σ : $s = 0$ if $d = 0$; otherwise $s = 1$ (fail-closed).

x : output · a : anchor · σ : anchor uncertainty



Tests pin 3σ at 0.50 and 9σ at 0.75.

Score boundaries: consistent < 0.50; tension < 0.75.

This score is an uncalibrated heuristic: it measures disagreement with one anchor, not a cause. SAT-042 uses it in the execution harness.

7 What does a checked reference establish?

Know what was checked

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token **record** **source pixel**

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG(row, col) = DN(b) & Ed25519(o) & leaf in STH & STHgeo.qa prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact, cid; one flipped mantissa bit gives mu5x2cks... · still trusted: BLAKE3's and the verifier's code

4-5 cell binding hashed cell = token cell = the lat/long the derivation used · the cell rule

6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code

8-9 upstream the STAC item matches the scene; the COG pixel at col 909, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum

10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right

12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the root geo.qa co-signed · geo.qa is also run by Vortex AI

15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI, DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION
Checkable observation handoff
EO evidence that travels between agents, checked by the receiver down to the source pixel.
Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem adds typed EO references and a source re-read, which caught a wrong pixel that hashing, signatures, logs and recomputation had passed (panel 6).

4 What exactly is handed over?

A compact reference to an exact record

The receiver uses an 84-character reference to fetch the record and its evidence.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.

SOURCE FILES
named, not hashed
B08 B04
B04 + B08 COG: Planetary Computer 277.9 + 281.9 MB row 9443, col 909 DN 1900, 3502

RECORD · 1,115 B
12 fields as decoded; no signature inside
kind primary privacy public
cell defi.zb572.xoso.zb1ec bound L1 schema d24rgwlq...
band indices.ndvi bound L1 signed 777er3yi...
tslot 20721 (25 Sep) bound L1 signed 2026-09-28T09:06:56Z
value 0.4708994708994709 re-computable L2
confidence 0.95 fixed per scene class (SCL4 here), not a measured uncertainty
sources sentinel_s2_12a_...B08_10m.tif ; ...B04_10m.tif (named L3)
captured 2026-09-25T05:42:51Z
derivation sentinel2_12a_indices.ndvi@1 · EPSG 32643 scene S2A_MS12A_20260925T054251_R005_T43SFS... (named L3)
DNs 3502 / 1900 · offset -1000 · + 9 more args

ADDRESS
BLAKE3 of the record
oj5ceccile62uvm6hedpuk67cj2ppq7z33mtsuyffgaxbxyaa
changes if one bit of the record changes
leaf of a batch root

ATTESTATION + LOG
Ed25519 over a batch root key 777er3yi...
Merkle log, RFC 6962 style, BLAKE3
log entries

emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk67cj2ppq7z33mtsuyffgaxbxyaa
handed over: 84 characters, 46 tokens (cl100k)

BLAKE3 identifies the 1,115-byte record. Its fields name the source files, sampled point and derivation; a batch signature covers the record address. Source-file hashes are absent from this record.

THE MEMORY, AS CODED · emem 18adb67 · fact.rs, cbor.rs, geo.rs, recall.rs, attest.rs, translog.rs, emem-attest

```
fact f = {kind, cell, band, tslot, value, unit, confidence, sources, derivation}fn key, args {, privacy, schema_cid, signer, signed_at} b = CBOR(f), declaration order, NaN as -0 as +0; fact_cid = base32(BLAKE3-256(b)); token emem:fact:cell:fact_cid; cell lat, q = round((lat+90)/180*(211-1)), lng, q = round((lng+180)/360*(27-1)); pitch 8.583e-5° = 9.52 × 8.06 m here recall tslot = floor(unix / 86 400) (20721 = 25 Sep 2026); key {cell, band, tslot}; as of: argmax {signed_at, cid} over signed_at ≤ t sign R = Merkle(sorted fact_cids); attestation = Ed25519(PreimageV1("attestation"){R, registry, schema}); a fact carries no signature log leaf = BLAKE3(0x00 || BLAKE3(CBOR(attestation))), node = BLAKE3(0x01 || 1 || i); RFC 6962 shape receipt sig = Ed25519(BLAKE3("emem.preimage.v1@0" || request_id || served_at || primitive || cells || fact_cids || merkle))
```

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve **Re-hash** **Bind** **Recompute** **Re-read**

agent bars: three Claude models pooled, 1 Oct 2026

	prose	JSON	RAG	opaque id	emem checks	first check that refuses
B acted on corrupted evidence	unaffected	refused	not applicable			
M2 seen in production	the only check that stops it	refused				
G0	nothing altered: accepted everywhere, never refused					
value						
M1	stated value moved by 1 ULP	12/12	12/12	9/12		C L0
M2	stated value rounded to "0.47"	6/12	6/12	0/12		C L0
M8	value 0.45, re-encoded and re-hashed	12/12	6/12	6/12	F	F L0
cell						
M9	record cited for another cell	7/12	0/12	6/12	E	E L1
M4	cell changed inside the record, re-hashed	7/12	11/12	6/12	F	F L0
time						
M10	tslot changed to look current, re-hashed	12/12	6/12	6/12	F	F L0
band						
M6	a record for another band handed over	12/12	0/12	6/12	E	E L1
source						
M11	source scene id changed, re-hashed	12/12	12/12	12/12	F	F L0
derivation						
M12	offset changed, value recomputed, re-hashed	12/12	11/12	8/12	F	F L0
M14	value (0.46) disagrees with the signed DNs	12/12	6/12	6/12	H	H L2
stale / current						
M5	an older record handed over as current	12/12	1/12	1/12	E	E L1
M16	a second signed version shown only to B	12/12	6/12	6/12	G	G L0
signature / id						
M3	1 ULP changed inside the served bytes	12/12	12/12	12/12	D	D L0
M7	token misquoted by one character	n/a	n/a	0/12		C L0
M13	value 0.45 under the forger's own key	12/12	6/12	6/12	F	F L0
source pixel						
M15	DNs read from the pixel 10 m south	12/12	12/12	12/12	I	I L3
entity						
M17	same record; A meant another entity (out of scope)					

B acts on corrupted evidence

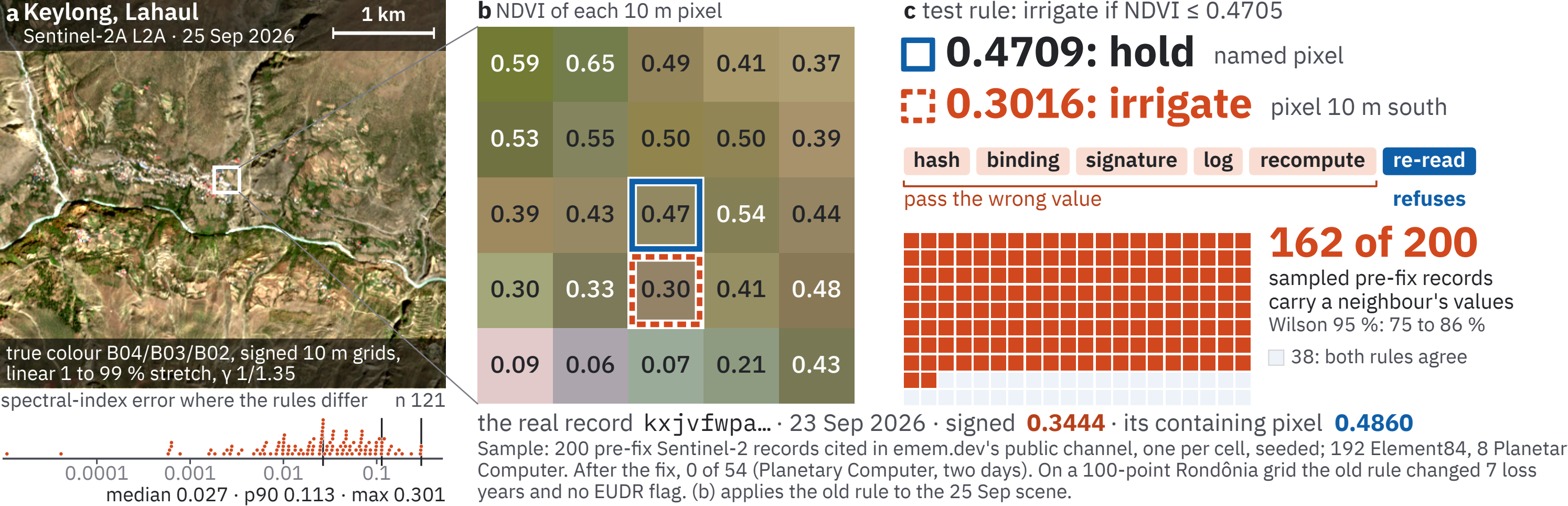
C resolve L0 **D** hash L0 **E** binding L1 **F** signature L0 **G** log L0 **H** recompute L2 **I** re-read L3 **D** to I are cumulative: each adds one check
Bars: agents' false acceptance, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.1, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 to 25 items, 1 Oct 2026.
Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Deeper checks expose different corruptions. Checked-reference agents made the expected decision on 71 of 72 genuine controls; one was refused (pooled Claude).

Each agent bar is 12 runs of one item; totals pool 23 items per condition (24 for the opaque id, 25 for emem). JSON and an opaque id caught some changes from their fields. On the live pre-fix record, agents acted 4 of 4 times uninstructed, 2 of 4 with resolve-and-bind instructions and 0 of 4 with a source re-read.

6 A valid record can preserve a wrong source read

NDVI = (DN₈ − DN₄) / (DN₈ + DN₄ + 2o), o = −1000 (product offset)



12 Use emem. Carry the evidence forward.

Separately, the same record and value came back through 11 client paths, with receipt signatures checked on 9 (30 Sep 2026). Listings show availability; panel 1 measures agent decisions.

ChatGPT · @emem plugin
“Signed facts about real places”
Enable emem; paste a token.

Claude · Directory plugin / MCP
“hand another agent a token it can check offline”
Resolve from chat or terminal.

Dify · Marketplace plugin
“Shared memory for AI agents. One address per fact, one signature you check.”
Referent Lock / EUDR workflows

GitHub MCP Registry · Server listing
“Satellites for AI.”
Install MCP server

ClawHub · OpenClaw plugin listing
“Shared, signed and content-addressed memory of the physical world...”
Install the OpenClaw plugin.

CONNECT

Setup · EO tasks
Reproduce

MORE CLIENTS
Cursor · VS Code · Gemini CLI

PROTOCOLS
MCP · A2A
REST / OpenAPI

DEVELOPER
Python · TypeScript · Docker
LangGraph memory · GitHub source

FRAMEWORK EXAMPLES
LangChain · LlamaIndex · AutoGen
CrewAI · Mastra · Agno (see setup notes)

DISCOVERY / MIRRORS
MCP Registry · MuleSoft Exchange · Glama · Smithery
Context7 · Hugging Face (older mirror)